

GenAI Prompt — Filter Products by Category

Prompt Used

Act as a Backend Architect. Using Node.js and Express, following the Controller-Route-Service pattern and this Contract:

Endpoint : GET /api/products

Query Param : category (string, optional, case-insensitive)

Success Response : 200 JSON array of products filtered by category

Error Response : 500 { message: "Unable to load products." }

Data Source : local JSON file (data/products.json)

And this Sequence:

1. TRIGGER — User clicks a category (e.g. "Hat") on the Frontend.
2. REQUEST — Browser sends GET /api/products?category=Hat to the server.
3. PROCESSING — Server (Gatekeeper) reads query param, fetches all products from JSON file, filters where product.category matches (case-insensitive).
4. RESPONSE — Server sends back a JSON array (Package) with status 200, or status 500 on failure.

Write:

- Backend/routes/products.js (Express Router, GET /)
- Backend/controllers/productsController.js (read req.query.category, call service)
- Backend/services/productsService.js (read JSON, filter by category)

Rules:

- Add comments explaining the Gatekeeper logic.
- Accept a query parameter for 'category'.
- If category is omitted, return all products.
- Matching is case-insensitive.
- Ensure the code has comments explaining how it works.

Result Summary

The prompt produced the three-layer structure below:

```
| File | Role |
|-----|-----|
| `routes/products.js` | Defines `GET /` route !' delegates to controller |
| `controllers/productsController.js` | Reads `req.query.category`, calls service, sends JSON |
| `services/productsService.js` | Reads `products.json`, applies category filter (Gatekeeper) |
```